

- Scaling down of images using HTML tags should be avoided. There is no point in using a bigger image and resizing it using the width and height attributes of the `<img>` tag.
- When using *Data URI*, the contents can be given in *inline* mode. This can be done for smaller sized images. So, instead of the following command...

```
.icon-test { background-image: url('test.png'); }
```

...use the code given below:

```
.icon-test { background-image: url('data:image/png;base64,iVBORw0KGgoAAAANSUuEugAAAAEAAAABQMAAA121bKAAAAA1BMEUUAACneJ3aAAAAARST1MAQ0BYZgAAAApJRE FUCNdjYAAAAIAAeIhVDMAAAAUuORK5CYII3D'); }
```

- Images generally contain data that are not required in Web usage. For example, the EXIF metadata can be stripped before uploading to the server.
- There are many tools to help you optimise images, such as TinyPNG, Compressor.io, etc. There are command line based tools also, such as jpegtran, imgopt, etc.

Performance analysis tools

There are many tools available to analyse the performance of Web pages. Some of these tools are illustrated in Figure 2.

There are component-specific tools, too. For example, for benchmarking JavaScript, the following tools may be used:

- JSPerf
- Benchmark.js
- JSlimus
- Matcha
- Memory-stats.js

For PHP, tools such as PHPBench and php-bench could be harnessed.

Minifiers

As stated earlier, minifying is one of the optimisation techniques, for which there are many tools. For HTML, the following Minifiers could be tried out:

- HTMLCompressor
- HTMLMinifier
- HTML_press
- Minimize

Some of the tools used for Minifying JavaScript and CSS are listed below:

- Uglifyjs2
- CSSmin.js
- Clean-css
- JShrink
- JSCompress
- YUI Compressor

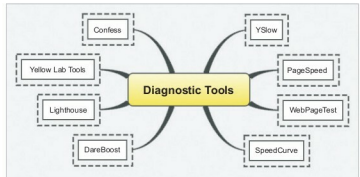


Figure 2: Performance analysis tools

Table 1

Tool	Description
Apache JMeter	This load testing tool is popular in the Java community.
Locust	This load testing tool can be used to specify user behaviour with Python. The capability to handle millions of simultaneous user requests can be tested with this tool.
Wrk	This is an HTTP benchmarking tool.
HTTPPerf	Different types of HTTP workloads shall be generated and tested. There are various ports available for HTTPPerf: - HTTPPerf.rb: Ruby interface - HTTPPerf.py: Python Port - HTTPPerf.js: JavaScript Port - Gohhttp: Go port

Benchmarking Web servers

Benchmarking of Web servers is an important mechanism in Web page/application optimisation. Table 1 provides a sample list of tools available for benchmarking Web servers.

The Web optimisation domain is really huge. This article has just provided a few start-up pointers, using which interested developers can proceed further in understanding the advanced technicalities of the topic. 

References

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